📘Complete MySQL Workbook (Beginner → Advanced)

This workbook is designed to **fully strengthen your MySQL concepts**. It is **topic-wise**, **progressive**, and **exam + interview oriented**. If you solve all questions honestly, your SQL fundamentals + advanced logic will become rock solid.

✂️Basics of Databases & MySQL

Conceptual Questions

1. What is a database?
2. Difference between DBMS and RDBMS.
3. What is MySQL?
4. Why is MySQL widely used?
5. What is a table, row, and column?
6. What is a schema?
7. Difference between SQL and MySQL.
8. What are metadata and data dictionary?
9. What is normalization and why is it needed?
10. What are the different types of databases?

Practical Tasks

1. Create a database for a college management system.
2. Show all databases.
3. Select a database.
4. Drop a database safely.

✒️Data Types in MySQL

Theory

1. What are numeric data types in MySQL?
2. Difference between INT, BIGINT, and TINYINT.
3. What is the difference between CHAR and VARCHAR?
4. When should TEXT be used?
5. Difference between DATE, DATETIME, TIMESTAMP.
6. What is ENUM? When should it be avoided?
7. What is DECIMAL and why is it preferred over FLOAT for money?

Practice

1. Create a table using all major data types.
2. Insert sample data and verify storage behavior.
3. Test max size of VARCHAR.

🛏️Table Creation & Constraints

Theory

1. What is PRIMARY KEY?
2. Difference between PRIMARY KEY and UNIQUE.
3. What is FOREIGN KEY?
4. What is NOT NULL constraint?
5. What is DEFAULT constraint?
6. What is CHECK constraint?
7. Can a table have multiple primary keys?
8. What is composite primary key?

Practice

1. Create Student and Course tables with constraints.
2. Add constraints after table creation.
3. Drop constraints.
4. Explain constraint violations with examples.

🌂INSERT, UPDATE, DELETE

Questions

1. Difference between INSERT and INSERT IGNORE.
2. What is INSERT INTO SELECT?
3. Difference between DELETE and TRUNCATE.
4. What happens when DELETE has no WHERE clause?
5. What is safe update mode?

Practice

1. Insert multiple rows at once.
2. Update records using conditions.
3. Delete records using subqueries.
4. Rollback a DELETE operation.

🌡️ SELECT Queries (Core of SQL)

Basic Queries

1. Select all columns from a table.
2. Select specific columns.
3. Use WHERE with comparison operators.
4. Use BETWEEN, IN, NOT IN.
5. Use LIKE with %, _.
6. Difference between = and LIKE.

Practice

1. Fetch students with marks > 80.
2. Find employees whose name starts with 'A'.
3. Fetch records between two dates.

🎒ORDER BY, LIMIT, DISTINCT

Questions

1. What is ORDER BY?
2. Can we use multiple columns in ORDER BY?
3. What is LIMIT?
4. Difference between DISTINCT and GROUP BY.

Practice

1. Fetch top 5 highest salaries.
2. Fetch second highest salary.
3. Fetch unique cities from customers table.

🎓Aggregate Functions

Theory

1. What are aggregate functions?
2. Difference between COUNT(*) and COUNT(column).
3. What is AVG?
4. What is MIN and MAX?

Practice

1. Find total salary of all employees.
2. Find average marks per subject.

3. Count number of students per department.

🎩GROUP BY & HAVING

Theory

1. Why GROUP BY is used?
2. Difference between WHERE and HAVING.
3. Can HAVING be used without GROUP BY?

Practice

1. Find departments having more than 10 employees.
2. Find cities where avg salary > 50k.
3. Count students grade-wise.

🎽Joins (Very Important)

Theory

1. What is JOIN?
2. Types of JOINs in MySQL.
3. Difference between INNER JOIN and LEFT JOIN.
4. What is SELF JOIN?
5. What is CROSS JOIN?

Practice

1. Join Student and Course tables.
2. Fetch employees with their manager name.
3. Find customers without orders.
4. Implement all joins on same tables.

🕘Subqueries

Theory

1. What is a subquery?
2. Difference between correlated and non-correlated subquery.
3. Can subquery return multiple rows?

Practice

1. Find employee with max salary.

2. Find students scoring above average.
3. Use subquery in FROM clause.

✂️✂️Views

Questions

1. What is a VIEW?
2. Advantages of views.
3. Can we update data through views?

Practice

1. Create a view for top students.
2. Drop and replace a view.

✂️✒️Indexes

Theory

1. What is an index?
2. Types of indexes in MySQL.
3. When should indexes be avoided?
4. What is clustered vs non-clustered index?

Practice

1. Create index on email column.
2. Check query performance using EXPLAIN.

✂️🛏️Stored Procedures

Theory

1. What is a stored procedure?
2. Advantages and disadvantages.
3. Difference between procedure and function.

Practice

1. Create a procedure to fetch employee details.
2. Use IN, OUT parameters.

✂️🌂Functions

Questions

1. What is a function?
2. Difference between built-in and user-defined functions.

Practice

1. Create function to calculate bonus.
2. Use string functions (SUBSTRING, CONCAT).

✂️🌡️ Triggers

Theory

1. What is a trigger?
2. Types of triggers.
3. BEFORE vs AFTER trigger.

Practice

1. Create trigger to log deletes.
2. Prevent negative salary insertion.

✂️🎒Transactions

Theory

1. What is a transaction?
2. What is ACID?
3. What is COMMIT, ROLLBACK, SAVEPOINT?

Practice

1. Simulate money transfer.
2. Use savepoints.

✂️🎓CTEs & Recursive Queries

Questions

1. What is CTE?

2. Difference between CTE and subquery.
3. What is recursive CTE?

Practice

1. Generate numbers using recursive CTE.
2. Employee hierarchy query.

✂️🎩Window Functions (Advanced)

Theory

1. What are window functions?
2. Difference between GROUP BY and window functions.
3. ROW_NUMBER vs RANK vs DENSE_RANK.

Practice

1. Find second highest salary using window function.
2. Running total of sales.

✂️🎽 Performance & Optimization

Questions

1. How to optimize SQL queries?
2. What is query execution plan?
3. What causes slow queries?

Practice

1. Optimize a slow query using indexes.
2. Compare subquery vs join performance.

✒️🏮 Real-World Case Studies

1. Design database for e-commerce system.
2. Write queries for order analytics.
3. Monthly revenue report.
4. User retention query.
5. Inventory management queries.
6. Banking transaction queries.

🔗Final Challenge

1. Write 10 complex queries using joins + subqueries.
2. Optimize each query.
3. Explain queries in plain English.
4. Convert business problem into SQL.
5. Debug incorrect SQL queries.

🕗Completion Goal

If you complete **all 145 questions**, you will be: - Interview ready - Exam confident - Strong in SQL logic - Comfortable with advanced MySQL

👉You can now ask me to: - Add **answers + explanations** - Convert this into **PDF / printable notes** - Add **LeetCode / interview-mapped questions** - Create **daily SQL practice plan**